DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I
WEEK 1: LECTURE NOTE

FUNCTIONS

Recall the following definition of Cartesian product of two sets:

Definition 1. *Suppose A and B are sets. The Cartesian product of A and B, denoted by $A \times B$, is the set which contains every ordered pair whose first coordinate is an element of A and second coordinate is an element of B; that is,*

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}.$$

Example 2. *If $A = \{1, 2, 3, 4\}$ and $B = \{5, 10\}$, then*

$$A \times B = \{(1, 5), (1, 10), (2, 5), (2, 10), (3, 5), (3, 10), (4, 5), (4, 10)\}.$$

Definition 3 (Equality of ordered Pairs)**.** *Two ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$. That is,* $\boxed{(a, b) = (c, d) \quad \Leftrightarrow \quad a = c \text{ and } b = d.}$

Many everyday phenomena involve two quantities that are related to each other by some rule of correspondence. The mathematical term for such a rule of correspondence is a relation. If A and B are sets, any subset of $A \times B$ is called a **relation** from A into B.

Definition 4 (Definition of a Function)**.** *A function f from a set A to a set B, denoted $f : A \to B$, is a relation that assigns to each element x in the set A exactly one element y in the set B. The set A is the domain (or set of inputs) of the function f, and the set B contains the range (or set of outputs).*

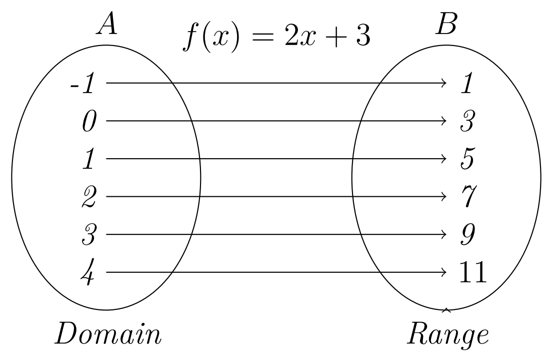
Function Notation: $y = f(x)$

1. x is the **independent variable** (**input value**)
2. y is the **dependent variable** (**output value**)
3. $f(x)$ is the value of the function at x.

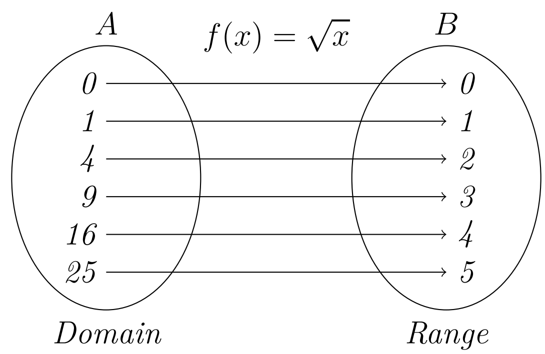
Characteristics of a Function f from Set A to Set B

1. Each element of A must be matched with an element of B.
2. Some elements of B may not be matched with any element of A.
3. Two or more elements of A may be matched with the same element of B
4. An element of A (the domain) cannot be matched with two different elements of B.

Example 5. ***Example of a Function***



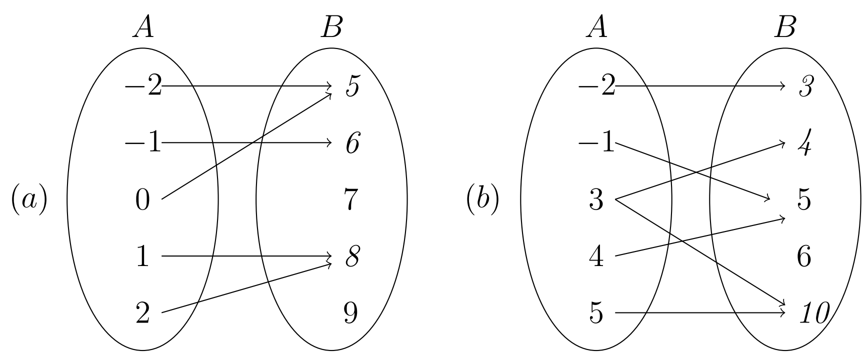
Example 6. ***Example of a Function***



Example 7. *If* $f(x) = \sqrt{x}$*, then*

$$\begin{aligned} &\text{Domain of } \; f = Dom(f) = [0, \infty). \\ &\text{Rang of } \; f = Ran(f) = [0, \infty). \end{aligned}$$

Example 8 (Testing for Functions)**.** *Does the relation describe a function? Explain your reasoning.*



In algebra, it is common to represent functions by equations or formulas involving two variables. For instance, the equation

$$y = x^2$$

represents the variable y as a function of the variable x. In this equation, x is the **independent variable** and y is the **dependent** variable.

- **Domain**: The domain of a function is the set of all values taken on by the independent variable x.

- **Range**: The range of a function is the set of all values taken on by the dependent variable y.

Example 9. ***Testing for Functions Represented Algebraically***. *Determine whether the equation represents y as a function of x.*

$$(a) \quad x^2 + y = 3$$
$$(b) \quad y^2 - x = 1$$

Solution. *To determine whether y is a function of x, try to solve for y in terms of x.*

(a) Solving for y

$x^2 + y = 1$	Write original equation.
$y = 1 - x^2$	Solve for y.

Each value of x corresponds to exactly one value of y. So, y is a function of x.

(b) Solving for y

$-x + y^2 = 1$	Write original equation.
$y^2 = x + 1$	Add x to each side.
$y = \pm\sqrt{x+1}$	Solve for y.

The $\pm$ indicates that for a given value of x there correspond two values of y. For instance, when $x = 3$, $y = 2$ or $y = -2$. So,y is not a function of x.

Function Notation. When an equation is used to represent a function, it is convenient to name the function so that it can be referenced easily. For example, you know that the equation $y = 1 - x^2$ describes y as a function of x. Suppose you give this function the name "f". Then you can use the following function notation.

Input	Output	Equation
x	$f(x)$	$f(x) = 1 - x^2$

Remark 10. ***Function Notation***

- ♣ *The symbol $f(x)$ is read as the value of f at x, or simply f of x.*
- ♣ *The symbol $f(x)$ corresponds to the y-value for a given x. So, you can write $y = f(x)$.*
- ♣ *Keep in mind that f is the name of the function, whereas $f(x)$ is the output value of the function at the input value x.*
- ♣ *In function notation, the **input is the independent variable** and the **output is the dependent variable**.*

Although f is often used as a convenient function name and x is often used as the independent variable, you can use other letters. For instance,

$$f(x) = x^2 - 4x + 7, \qquad f(t) = t^2 - 4t + 7, \qquad \text{and} \qquad g(s) = s^2 - 4s + 7,$$

all define the same function. In fact, the role of the independent variable is that of a "**placeholder**." Consequently, the function could be written as

$$f(\square) = (\square)^2 - 4(\square) + 7.$$

Example 11 (Evaluating a Function). *Let* $g(x) = -x^2 + 4x + 1$. *Find each value of the function.*

$$\begin{aligned} &(a)\quad g(2)\\ &(b)\quad g(t)\\ &(c)\quad g(x+2) \end{aligned}$$

Solution.

(a) Replacing x *with* 2 *in* $g(x) = -x^2 + 4x + 1$ *yields the following.*

$$g(2) = -(2^2) + 4(2) + 1 = -4 + 8 + 1 = 5.$$

(b) Replacing x *with* t *in* $g(x) = -x^2 + 4x + 1$ *yields the following.*

$$g(t) = -(t)^2 + 4(t) + 1 = -t^2 + 4t + 1.$$

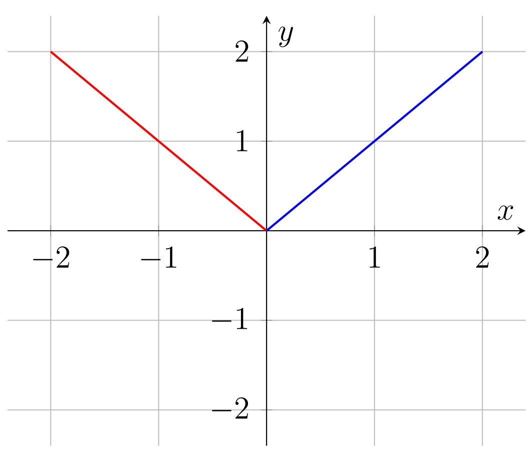
(c) Replacing x *with* $x + 2$ *in* $g(x) = -x^2 + 4x + 1$ *yields the following.*

$$\begin{aligned} g(x+2) &= -(x+2)^2 + 4(x+2) + 1 && \text{Substitute } x+2 \text{ for } x\\ &= -(x^2 + 4x + 4) + 4x + 8 + 1 && \text{Multiply}\\ &= -x^2 - 4x - 4 + 4x + 8 + 1 && \text{Distributive Property}\\ &= -x^2 + 5 && \text{Simplify.} \end{aligned}$$

Definition 12 (**A Piecewise-Defined Function**). *A function defined by two or more equations over a specified domain is called a* ***piecewise-defined function****.*

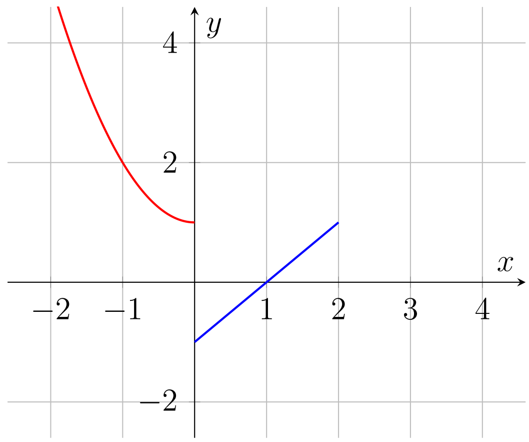
Example 13 (**The parent absolute value function** $f(x) = |x|$). *The function* $f(x) = |x|$ *can be written as a piecewise-defined function*

$$f(x) = |x| = \begin{cases} x, & x \geq 0\\ -x, & x < 0. \end{cases}$$



Example 14 (**A Piecewise-Defined Function**). *Evaluate the function when* $x = -1$ *and* $x = 0$.

$$f(x) = \begin{cases} x^2 + 1, & x < 0\\ x - 1, & x \geq 0. \end{cases}$$



Solution. *Because* -1 *is less than* 0*, use* $f(x) = x^2 + 1$ *to obtain*

$$f(-1) = (-1)^2 + 1 = 1 + 1 = 2.$$

Because 0 *is greater than* 0*, use* $f(x) = x - 1$ *to obtain*

$$f(0) = 0 - 1 = -1.$$

THE DOMAIN OF A FUNCTION

The domain of a function can be described explicitly or it can be implied by the expression used to define the function.

Definition 15 (Implied Domain)**.** *The implied domain is the set of all real numbers for which the expression is defined. Common type of implied domain:*

1. ***excludes*** x***-values that result in division by zero.***
2. ***avoids even roots of negative numbers.***

Example 16 (Finding the Domain of a Function)**.** *Find the implied domain of the function*

$$f(x) = \frac{1}{x^2 - 4}.$$

Solution. *Excluding* x*-values that yield zero in the denominator, the domain of* f *is the set of all real numbers except* ± 2*. These two values are excluded from the domain because division by zero is undefined.*

Example 17 (Finding the Domain of a Function)**.** *Find the implied domain of the function*

$$f(x) = \sqrt{x}.$$

Solution. *Excluding* x*-values that would cause result in the square root of a negative number, the domain of* f *is the interval* $[0, \infty)$*.*

Example 18 (Finding the Domain and Range of a Function)**.** *Find the domain and range of the function*

$$f(x) = \sqrt{9 - x^2}.$$

Solution. *To exclude x-values that would cause result in the square root of a negative number, the function f is defined only for x-values for which*

$$9 - x^2 \geq 0 \ , \qquad (3-x)(3+x) \geq 0.$$

By solving this inequality, we will find that the domain of f is

$$-3 \leq x \leq 3.$$

When the x-values extend from to -3 to 3 then the $y = f(x)$-values extend from 0 to 3. Therefore, the range of f is

$$0 \leq y \leq 3.$$

Definition 19. *For a function f, the ratio $\dfrac{f(x+h)-f(x)}{h}$, $h \neq 0$, is called a **difference quotient**.*

Example 20 (Application)**.** *For $f(x) = x^2 - 4x + 7$, find $\dfrac{f(x+h)-f(x)}{h}$.*
Solution.

$$\begin{aligned}
\frac{f(x+h)-f(x)}{h} &= \frac{[(x+h)^2 - 4(x+h) + 7] - [x^2 - 4x + 7]}{h} \\
&= \frac{[x^2 + 2xh + h^2 - 4x - 4h + 7] - [x^2 - 4x + 7]}{h} \\
&= \frac{x^2 + 2xh + h^2 - 4x - 4h + 7 - x^2 + 4x - 7]}{h} \\
&= \frac{2xh - 4h + h^2}{h} = 2x - 4 + h
\end{aligned}$$

SUMMARY OF FUNCTION TERMINOLOGY

1. **FUNCTION**
   A **function** is a relationship between two variables such that to each value of the independent variable there corresponds exactly one value of the dependent variable.

2. **FUNCTION NOTATION**
   $y = f(x)$

   (a) f is the name of the function.

   (b) y is the dependent variable, or output value.

   (c) x is the independent variable, or input value.

   (d) $f(x)$ is the value of the function at x.

3. **DOMAIN**
   The **domain of a function** is the set of all values (inputs) of the independent variable for which the function is defined. If x is in the domain of f, then f is said to be defined at x. If x is not in the domain of f, then f is said to be undefined at x.

4. RANGE
   The **range of a function** is the set of all values (outputs) assumed by the dependent variable (that is, the set of all function values).

5. **IMPLIED DOMAIN**
   If f is defined by an algebraic expression and the domain is not specified, then the **implied domain** consists of all real numbers for which the expression is defined. Common type of implied domain:

   (a) **excludes x-values that result in division by zero.**

   (b) **avoids even roots of negative numbers.**

The Graph of an Equation

Definition 21 (Solution Point of an Equation in Two Variables)**.** *For an equation in the variables x and y, a point (a, b) is a* ***solution point*** *when substitution of a for x and b for y satisfies the equation. Most equations in two variables have infinitely many solution points.*

Example 22 (Determining Solution Points)**.** *Give at least four solution points of the equation $3x + y = 5$.*
Solution.

- $(1, 2)$ *is a solution point of the equation, because substitution of* 1 *for x and* 2 *for y satisfies the equation.*

- $(-1, 0)$ *is* ***not*** *a solution point of the equation, because substitution of* -1 *for x and* 0 *for y gives the false statement* $-3 = 5$.

- $(-1, 8)$ *is a solution point of the equation, because substitution of* -1 *for x and* 8 *for y satisfies the equation.*

- $(-\frac{1}{3}, 6)$ *is a solution point of the equation, because substitution of* $-\frac{1}{3}$ *for x and* 6 *for y satisfies the equation.*

- $(2, 1)$ *is* ***not*** *a solution point of the equation, because substitution of* 2 *for x and* 1 *for y gives the false statement* $7 = 5$.

- $(\frac{1}{2}, \frac{7}{2})$ *is a solution point of the equation, because substitution of* $\frac{1}{2}$ *for x and* $\frac{7}{2}$ *for y satisfies the equation.*

Definition 23. ***(The Graph of an Equation and Intercepts)***

1. *The set of all solution points of an equation is called* ***the graph of the equation****.*

2. *The points at which a graph touches or crosses an* ***axis*** *are called* ***the intercepts of the graph****.*

3. *A point at which a graph touches or crosses an* ***x-axis*** *is called an* ***x-intercept of the graph****.*

4. *A point at which a graph touches or crosses a* ***y-axis*** *is called an* ***y-intercept of the graph****.*

Sketching the Graph of an Equation by Point Plotting

1. If possible, rewrite the equation so that one of the variables is isolated on one side of the equation.
2. Make a table of values showing several solution points.
3. Plot these points on a rectangular coordinate system.
4. Connect the points with a smooth curve or line.

Example 24 (Sketching a Graph by Point Plotting)**.** *Use point plotting and graph paper to sketch the graph of* $3x + y = 6$.

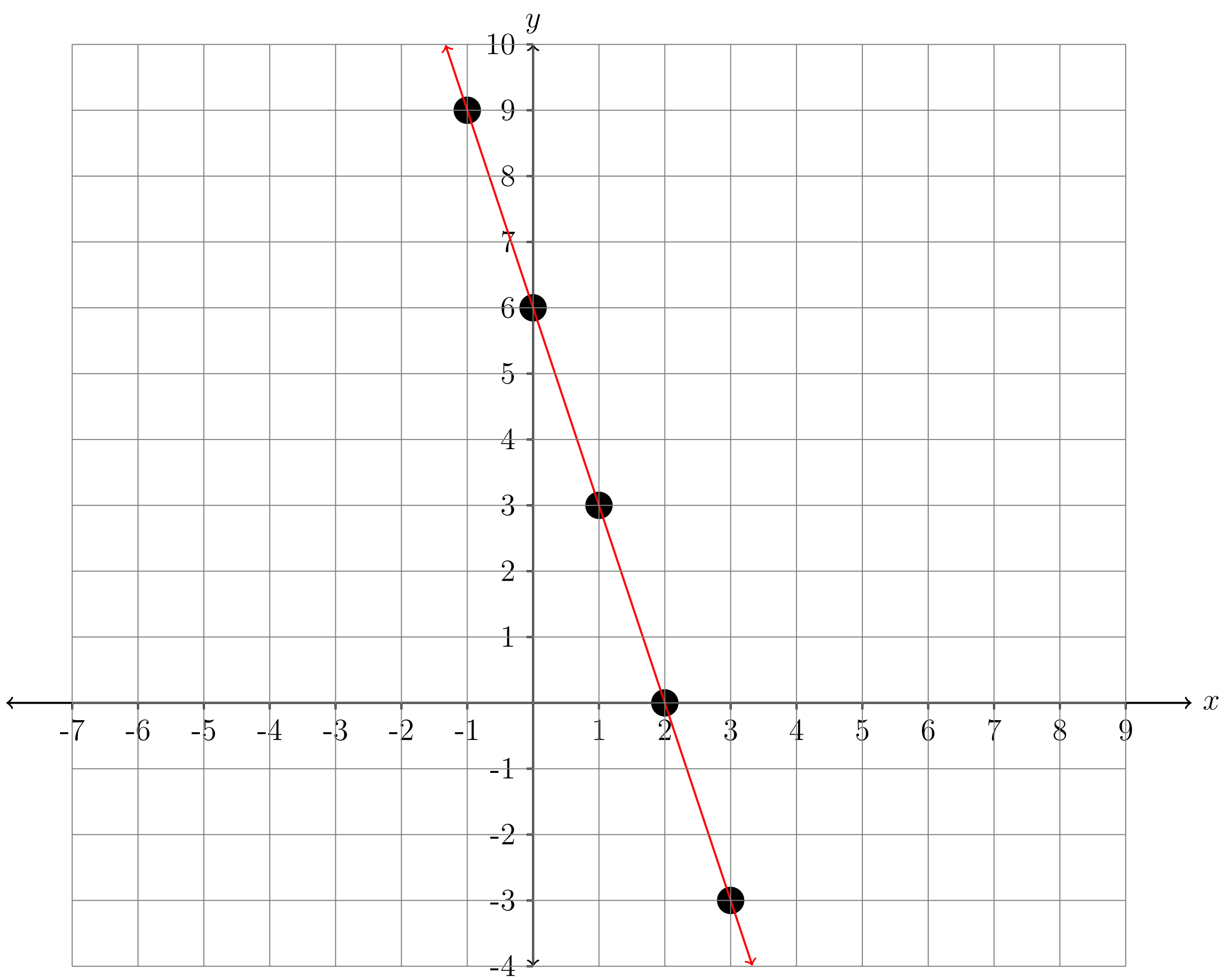
Solution. In this case we can isolate the variable y. The equation is rewritten as

$$y = 6 - 3x.$$

Using negative and positive values of x, and $x = 0$, we can obtain the following table of values (solution points).

x	-1	0	1	2	3
$y = 6 - 3x$	9	6	3	0	-3
Solution Point	$(-1, 9)$	$(0, 6)$	$(1, 3)$	$(2, 0)$	$(3, -3)$

Next, plot the solution points and connect them, as shown in the Figure below. It appears that the graph is a straight line. We note that **the point** $(0, 6)$ **is the y-intercept of the graph** and **the point** $(2, 0)$ **is the x-intercept of the graph**.



Example 25 (Sketching a Graph by Point Plotting)**.** *Use point plotting and graph paper to sketch the graph of*

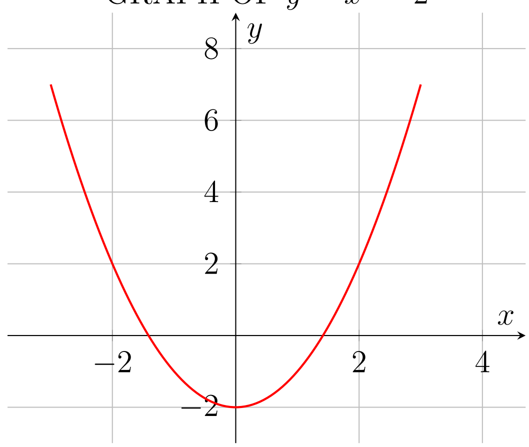
$$y = x^2 - 2.$$

Solution. *Because the equation is already solved for y, make a table of values by choosing several convenient values of x and calculating the corresponding values of y.*

x	-2	-1	0	1	2	3
$y = x^2 - 2$	2	-1	-2	-1	2	7
Solution Point	$(-2, 2)$	$(-1, -1)$	$(0, -2)$	$(1, -1)$	$(2, 2)$	$(3, 7)$

Next, plot the solution points, as shown in the Figure below. Finally, connect the points with a smooth curve. This graph is called a ***parabola****. Note that the points $(-\sqrt{2}, 0)$ and $(\sqrt{2}, 0)$ are the x-intercepts of the graph; and the point $(0, -2)$ is the y-intercept of the graph.*

GRAPH OF $y = x^2 - 2$

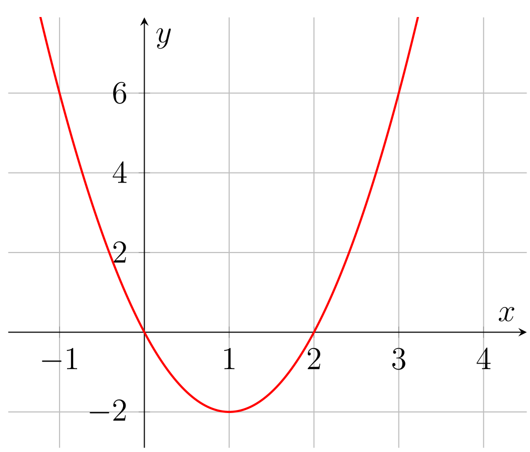


By the definition of a function, at most one y-value corresponds to a given x-value. It follows, then, that a vertical line can intersect the graph of a function at most once. This leads to the Vertical Line Test for functions.

Remark 26 (Vertical Line Test for Functions)**.** *A set of points in a coordinate plane is the graph of y as a function of x if and only if no vertical line intersects the graph at more than one point.*

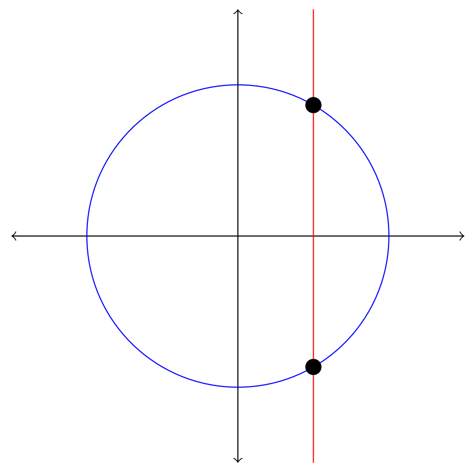
Example 27 (Vertical Line Test for Functions)**.** *Use the Vertical Line Test to decide whether the graph of the equation represent y as a function of x.*

(a) $y = 2x^2 - 4x$



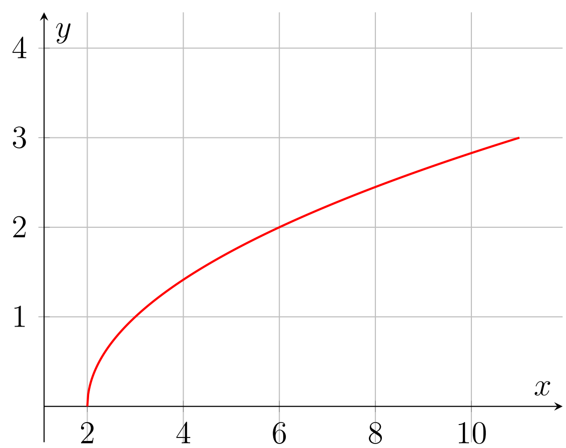
The graph of the equation represent y as a function of x.

(b) $x^2 + y^2 = 4$



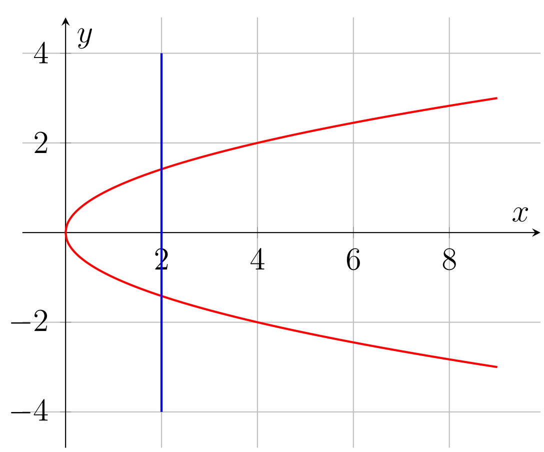
The graph of the equation does not represent y as a function of x because there is a vertical line that intersects the graph at two distinct points.

(c) $y = \sqrt{x-2}$



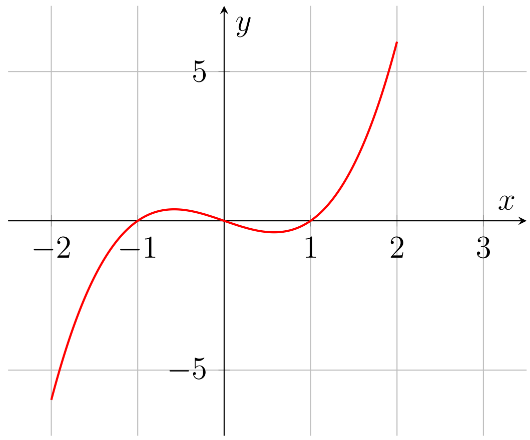
The graph of the equation represents y as a function of x.

(d) $y^2 = x$



*The graph of the equation **does not** represent y as a function of x because there is a vertical line that intersects the graph at two distinct points.*

(e) $y = x^3 - x$



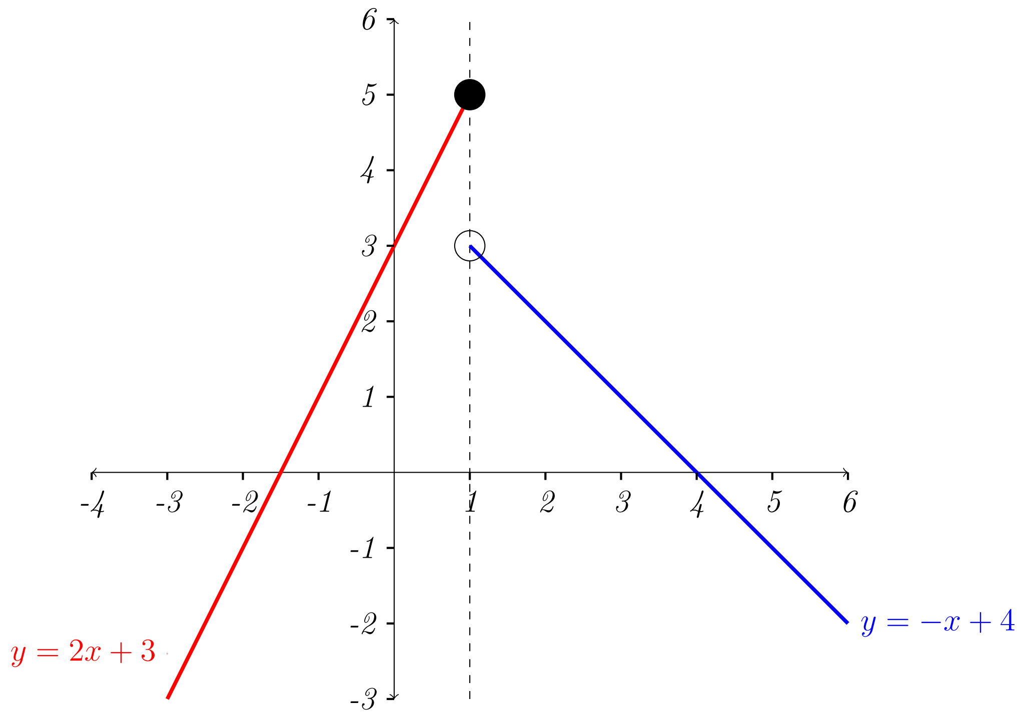
The graph of the equation represents y as a function of x.

(f) ***SKETCHING A PIECEWISE DEFINED FUNCTION***

(i) *Sketch the graph of*

$$f(x) = \begin{cases} 2x+3, & x \le 1 \\ -x+4, & x > 1. \end{cases}$$

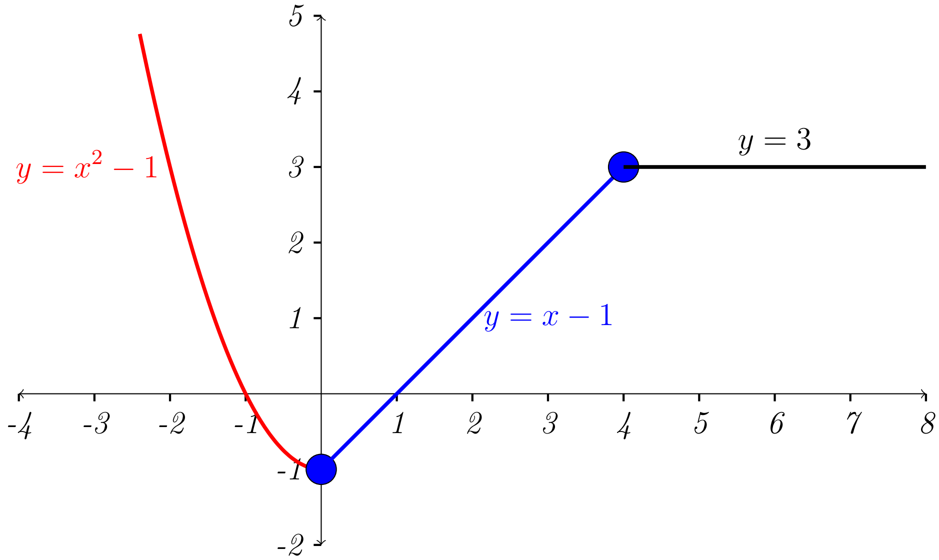
Solution. *The piecewise defined function f is composed of two linear functions joint at $x = 1$.*



(ii) Sketch the graph of

$$f(x) = \begin{cases} x^2 - 1, & x < 0 \\ x - 1, & 0 \leq x \leq 4 \\ 3, & x > 4. \end{cases}$$

Solution. *The piecewise defined function f is composed of three functions:*



Types of Functions

1. One-to-one (injective) function
   A one-to-one function maps each element of the domain to a unique element of the range, ensuring that no two different inputs produce the same output. In a one-to-one function $f : A \to B$, given any $y \in B$ there is at most one $x \in A$ that can be paired with y.

   A graph of a function can also be used to determine whether a function is one-to-one using the horizontal line test:

   If each horizontal line crosses the graph of a function at no more than one point, then the function is one-to-one.

2. Onto (surjective) function
   An onto function is such that for every element in the codomain there exists an element in domain which maps to it. Mathematically stated as

   A function $f : A \to B$ is called **onto** if for all $b \in B$ there is an $a \in A$ such that $f(a) = b$. An onto function maps every element in the range to at least one element in the domain, guaranteeing that no elements are left unmapped in the range.

3. One-to-one correspondence (bijective function)
   We can say that a function $f : A \to B$ is said to be a **bijective function** or **bijection** or **1-1 correspondence** if f is both one-to-one (injective) and onto (surjective).